ASESII 24A02 Final Project - Part 2

2-way ANOVA method

Group 02
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Contents

1 Abstract

Reproducible setup

```
required_packages <- c("tidyverse", "glmnet", "broom", "knitr", "car", "ggplot2")
missing_packages <- required_packages[
  !vapply(required_packages, requireNamespace, logical(1), quietly = TRUE)
]
if (length(missing_packages)) {
  stop("Install the required packages before rendering: ",
       paste(missing_packages, collapse = ", "))
}

library(tidyverse)
library(glmnet)
```

```r
library(broom)
library(knitr)
library(car)
library(ggplot2)
# `params` exists only while knitting; the fallback
# keeps interactive chunk runs working.
has_params <- exists("params") && !is.null(params$group_number)
group_number <- if (has_params) as.integer(params$group_number) else 4L
if (length(group_number) != 1L || is.na(group_number) || group_number < 1L) {
  stop("Set params$group_number to your assigned positive group number.")
}

seeds <- list(
  split = 240200L + group_number,
  cv = 240300L + group_number,
  features = 240400L + group_number
)
```

Shared helper functions

```r
find_exam_data <- function(filename) {
  input <- knitr::current_input(dir = TRUE)
  input_dir <- if (length(input) && nzchar(input)) dirname(input) else getwd()
  candidates <- c(
    file.path(input_dir, "data", filename),
    file.path(input_dir, "..", "data", filename),
    file.path(getwd(), "data", filename)
  )
  existing <- candidates[file.exists(candidates)]
  if (!length(existing)) stop("Cannot locate ", filename, " by a relative path.")
  hits <- unique(normalizePath(existing, winslash = "/", mustWork = TRUE))
  if (length(hits) > 1L && length(unique(unname(tools::md5sum(hits)))) > 1L) {
    stop("Multiple nonidentical copies of ", filename, " were found.")
  }
  hits[[1L]]
}

split_rows <- function(n, train_prop = 0.80, seed) {
  stopifnot(n >= 10L, train_prop > 0, train_prop < 1)
  set.seed(seed)
  train <- sort(sample.int(n, floor(train_prop * n), replace = FALSE))
  list(train = train, test = setdiff(seq_len(n), train))
}

fit_scaler <- function(x, tol = 1e-12) {
  x <- as.matrix(x)
```

```r
  center <- colMeans(x)
  centered <- sweep(x, 2L, center, "-")
  scale <- sqrt(colMeans(centered^2))
  keep <- is.finite(scale) & scale > tol
  if (!any(keep)) stop("No nonconstant training predictor remains.")
  list(
    center = center[keep],
    scale = scale[keep],
    keep = keep,
    dropped = colnames(x)[!keep]
  )
}

apply_scaler <- function(x, prep) {
  x <- as.matrix(x)[, prep$keep, drop = FALSE]
  x <- sweep(x, 2L, prep$center, "-")
  sweep(x, 2L, prep$scale, "/")
}

make_foldid <- function(n, k = 5L, seed) {
  if (k < 3L || k > n) stop("Require 3 <= k <= number of training rows.")
  set.seed(seed)
  sample(rep(seq_len(k), length.out = n))
}

fit_cv_glmnet <- function(x, y, alpha, foldid) {
  glmnet::cv.glmnet(
    x = x,
    y = y,
    family = "gaussian",
    alpha = alpha,
    foldid = foldid,
    standardize = FALSE,
    intercept = TRUE,
    type.measure = "mse"
  )
}

score_regression <- function(y, prediction) {
  y <- as.numeric(y)
  prediction <- as.numeric(prediction)
  if (length(y) != length(prediction) || any(!is.finite(c(y, prediction)))) {
    stop("Metrics require equal-length finite vectors.")
  }
  c(RMSE = sqrt(mean((y - prediction)^2)),
    MAE = mean(abs(y - prediction)))
}
```

```r
count_nonzero <- function(fit, s = "lambda.min", tol = 1e-8) {
  b <- as.matrix(stats::coef(fit, s = s))
  slopes <- b[rownames(b) != "(Intercept)", 1L]
  sum(abs(slopes) > tol)
}

safe_condition_numbers <- function(x, lambda, tol = 1e-10) {
  eigenvalues <- eigen(
    crossprod(x) / nrow(x),
    symmetric = TRUE,
    only.values = TRUE
  )$values
  largest <- max(eigenvalues)
  smallest <- max(min(eigenvalues), 0)
  cutoff <- tol * max(1, largest)
  c(
    gram = if (smallest <= cutoff) Inf else largest / smallest,
    ridge = (largest + lambda) / (smallest + lambda)
  )
}
analysis_model <- function(x, y, alpha, foldid) {
  model <- fit_cv_glmnet(x = x, y = y, alpha=alpha, foldid = foldid)
  list(
    model = model,
    lambda_min = model$lambda.min,
    mse_min = model$cvm[model$lambda == model$lambda.min],
    cond_min = safe_condition_numbers(x, lambda = model$lambda.min)[2],
    coef_min = coef(model, "lambda.min"),
    coef_min_norm = sqrt(sum(coef(model, "lambda.min")[-1]^2)),
    nonzero_min = count_nonzero(model, s="lambda.min"),
    lambda_1se = model$lambda.1se,
    cond_1se = safe_condition_numbers(x, lambda = model$lambda.1se)[2],
    mse_1se = model$cvm[model$lambda == model$lambda.1se],
    coef_1se = coef(model, "lambda.1se"),
    coef_1se_norm = sqrt(sum(coef(model, "lambda.1se")[-1]^2)),
    nonzero_1se = count_nonzero(model, s="lambda.1se")
  )
}
```

2 Dataset selection

```r
data <- read.csv("data/student_performance.csv", header = FALSE)
colnames(data) <- c(
  "student_id",
  "gender",
```

```
  "study_time_hours",
  "attendance_percent",
  "sleep_hours",
  "parental_education",
  "internet_access",
  "extracurricular_activities",
  "part_time_job",
  "previous_grade",
  "final_exam_score",
  "final_grade"
)

config <- list(
  file = "student_performance.csv",
  response = "final_exam_score",
  excluded = "final_exam_score"
)
data_raw <- read.csv(find_exam_data(config$file), check.names = FALSE)
predictors <- setdiff(names(data_raw), config$excluded)
analysis_data <- data_raw[, c(config$response, predictors), drop = FALSE]

if (anyNA(analysis_data)) stop("Declare a training-only missing-value plan.")
```

Table 1: Response and candidate predictors, with the integrity checks run before the split.

Variable	Role	Type	Distinct	Missing
student_id	Predictor	integer	1000	0
gender	Predictor	character	2	0
study_time_hours	Predictor	numeric	70	0
attendance_percent	Predictor	numeric	327	0
sleep_hours	Predictor	numeric	65	0
parental_education	Predictor	character	5	0
internet_access	Predictor	character	2	0
extracurricular_activities	Predictor	character	2	0
part_time_job	Predictor	character	2	0
previous_grade	Predictor	numeric	434	0
final_exam_score	Response	numeric	342	0
final_grade	Predictor	character	5	0

For 2-Way ANOVA, this method requires the dataset to have at least 2 categorical variables along with a continuous response variable for the analysis.

In general, from our chosen dataset, we decide to evaluate the factors that directly affect the student performance on the score. This data constains information over 1000 students, including the daily habits (the study time, hours of sleep, internet access), lifestyles (extracurricular activities, parttime job) demographic information (gender, education level of their parents) and academic performance (final exam score, final grade). It's ideal for exploratory data analysis and 2-Way ANOVA analysis based on the behavioral and demographic factors.

Specifically, there are 6 categorical factors: `gender` (Male/Female), `parental_education`(PhD/Master/Bachelors/ School/None), `internet_access`(Yes/No), `extracurricular_activities`(Yes/No), `part_time_job`(Yes/No), `final_grade`(A/B/C/D/F). In this method, we pick the continuous response on `final_exam_score` to test the hypothesis whether any factors that affect this response as well as any relationships/interactions between 2 categorical factors.

3 Research Questions & Hypotheses

We investigate whether students' participation in extracurricular activities and part-time jobs affects their final exam scores, and whether the two factors interact with each other.

3.0.1 Research Question 1: Extracurricular Activities

Does participation in extracurricular activities affect students' final exam scores?

- H_0: The mean final exam score is the same for students who participate in extracurricular activities and those who do not.

$$H_0 : \mu_{\text{Yes}} = \mu_{\text{No}}$$

- H_A: The mean final exam score differs between students who participate in extracurricular activities and those who do not.

$$H_A : \mu_{\text{Yes}} \neq \mu_{\text{No}}$$

3.0.2 Research Question 2: Part-time Job

Does having a part-time job affect students' final exam scores?

- H_0: The mean final exam score is the same for students with and without a part-time job.

$$H_0 : \mu_{\text{Job}} = \mu_{\text{No Job}}$$

- H_A: The mean final exam score differs between students with and without a part-time job.

$$H_A : \mu_{\text{Job}} \neq \mu_{\text{No Job}}$$

3.0.3 Research Question 3: Interaction Effect

Does the effect of having a part-time job on final exam scores depend on whether students participate in extracurricular activities?

- H_0: There is no interaction effect between extracurricular activities and part-time jobs.

$$H_0 : (\alpha\beta)_{ij} = 0 \quad \text{for all } i, j$$

- H_A: There is an interaction effect between extracurricular activities and part-time jobs.

$$H_A : (\alpha\beta)_{ij} \neq 0 \quad \text{for at least one } i, j$$

```
##   student_id gender study_time_hours attendance_percent sleep_hours
## 1 student_id gender study_time_hours attendance_percent sleep_hours
## 2          1   Male              4.0               98.0         6.5
## 3          2 Female              6.3              100.0         5.7
## 4          3   Male              4.9               85.3         7.9
## 5          4   Male              2.6               77.5         8.0
## 6          5   Male              2.2               89.6         4.6
##   parental_education internet_access extracurricular_activities part_time_job
## 1 parental_education internet_access extracurricular_activities part_time_job
## 2          Bachelors             Yes                        Yes            No
## 3        High School             Yes                        Yes           Yes
## 4          Bachelors             Yes                         No           Yes
## 5               None             Yes                        Yes            No
## 6          Bachelors             Yes                         No           Yes
##   previous_grade final_exam_score final_grade
## 1 previous_grade final_exam_score final_grade
## 2           76.9            100.0           A
## 3           75.5            100.0           A
## 4           88.5             97.3           A
## 5           85.1             83.8           B
## 6           61.8             68.3           D
```

```
##  [1] "student_id"                 "gender"
##  [3] "study_time_hours"           "attendance_percent"
##  [5] "sleep_hours"                "parental_education"
##  [7] "internet_access"            "extracurricular_activities"
##  [9] "part_time_job"              "previous_grade"
## [11] "final_exam_score"           "final_grade"
```

```
##
##          Bachelors        High School            Masters               None
##                308                356                184                102
## parental_education                PhD
##                  1                 50
```

```
##
## extracurricular_activities                                    No
##                          1                                   428
##                        Yes
##                        572

##
##          A           B           C           D           F final_grade
##        284         354         261          89          12           1
```